

PROBLEM STATEMENT

- Acute Kidney Injury (AKI) affects about 1 in 3 ICU patients. It raises mortality, ICU stay and cost, and is usually detected only after kidney damage has begun.
- Existing ML models rely on thousands of features and large labeled datasets, which makes them hard to train, hard to explain and hard to transfer to other hospitals.

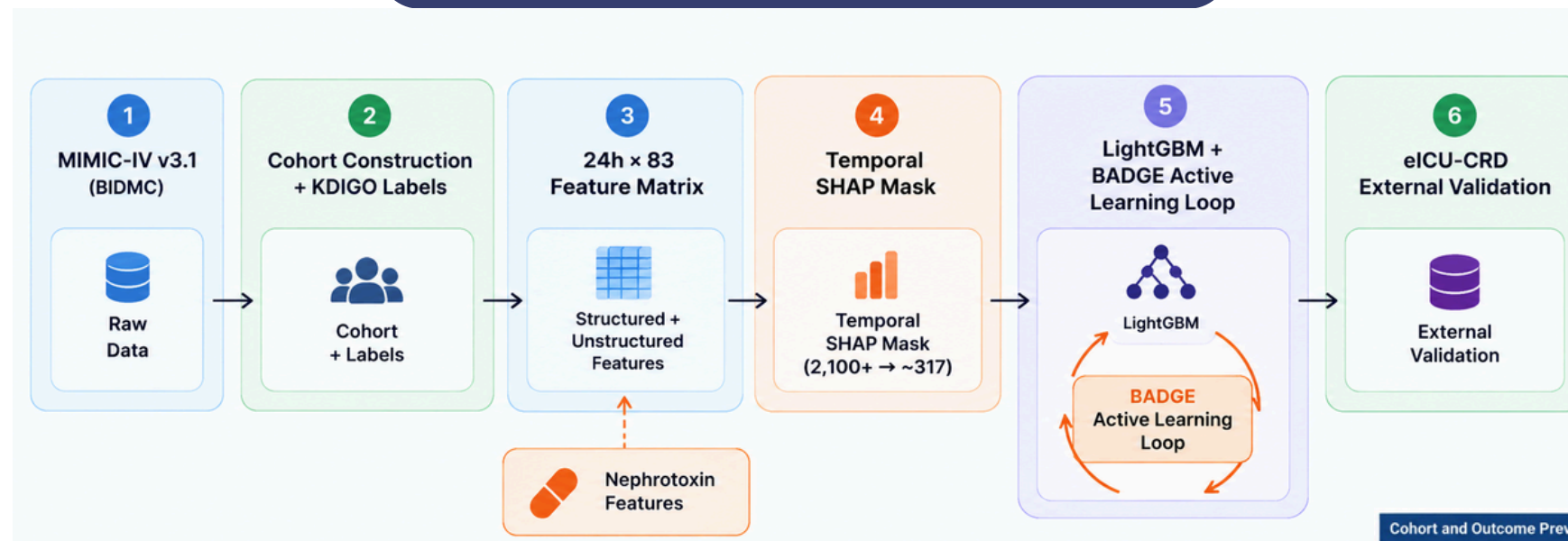
OBJECTIVE

- Reproduce a published AKI forecasting benchmark to validate our data pipeline.
- Compress ~2,100+ clinical features to ~317 without losing predictive power, using temporal SHAP as a preprocessing step.
- Cut labeling cost with SHAP-guided active learning (BADGE), targeting near-full performance with ~15% of labeled data.
- Capture drug-induced kidney risk through nephrotoxin time-series and renal trajectory features.
- Test generalization on eICU-CRD (208 US hospitals) with zero retraining.

PROPOSED METHODOLOGY

- Cohort and labels: ICU stays from MIMIC-IV, AKI labeled by KDIGO creatinine and urine-output criteria, with prior-AKI exclusion.
- Feature engineering: hourly time series, missingness flags, nephrotoxin and creatinine-slope features.
- Temporal SHAP mask: TreeSHAP values computed on a held-out calibration set rank every feature-hour. Low-impact features are dropped, so SHAP acts as a selector rather than just an explainer.
- Modeling: LightGBM trained on the compressed ~317-feature space.
- SHAP-BADGE active learning: BADGE picks the most informative patients to label, working inside the SHAP-compressed space.
- Validation: frozen model tested on eICU-CRD with no retraining.

SYSTEM ARCHITECTURE



DATASET AND FEATURES

I] Dataset

- MIMIC-IV v3.1, Beth Israel Deaconess Medical Center. Training and internal testing, ~[59,655] ICU stays, ~[12.8%] 2 & 3 stage AKI (KDIGO criteria), temporal 60/20/20 split.
- eICU-CRD v2.0, 208 US hospitals. External validation only.

II] Features

- Hourly 24-hour windows of vitals, labs, urine output, GCS and medications, plus clinical-note embeddings (BiomedBERT).
- Missingness flags computed before imputation, then forward-fill.
- Nephrotoxin features: 10 drug classes, multiple dimensions per hour across 24 h.
- Renal trajectory features: creatinine slope.
- Leakage-safe preprocessing: scalers are fit only on the training split.



Mahita G
Mentor



Tanusha Prakash
PES1UG24CS494



Aryan Revankar
PES1UG24CS084



Vennela Shakthi
PES1UG24CS525

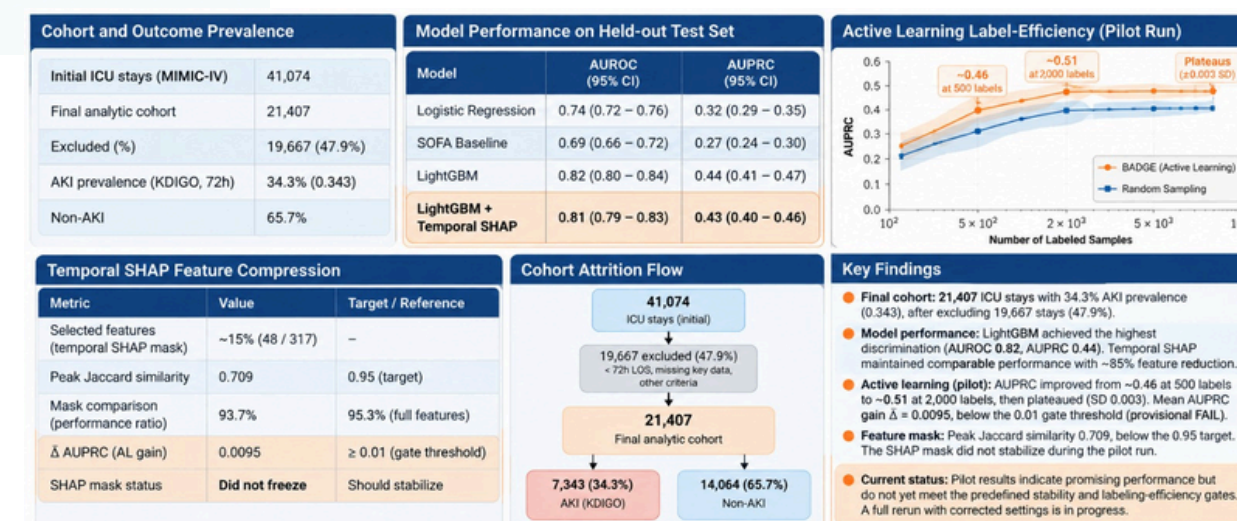


Manasvi D R
PES1UG24CS262

RESULTS AND DISCUSSION

- Pre-flight verification before real data caught data leakage, competing-risk logic errors and missingness-flag ordering bugs.
- A full pipeline audit fixed itemid errors (urine output, GCS), a non-temporal split caused by MIMIC-IV date-shifting, and a BADGE embedding issue.
- Evaluation metrics: AUROC, AUPRC, plus label efficiency curves for active learning.

RESULTS



CONCLUSION AND FUTURE WORK

- We reproduced a published AKI forecasting benchmark and extended it with three novel contributions.
- SHAP used as a preprocessing mask cuts the feature space by ~85%, which makes models lighter and more interpretable.
- Nephrotoxin trajectories add drug-exposure signal that standard feature sets miss.